A Game Recommender System Using Collaborative Filtering (GAMBIT)

[1*]Syed Muhammad Anwar, [1]Talha Shahzad, [1]Zunaira Sattar, [1]Rahma Khan, [2]Muhammad Majid

[1]Department of Software Engineering, FTIE, University of Engineering and Technology, Taxila, Pakistan
[2] Department of Computer Engineering, FTIE, University of Engineering and Technology, Taxila, Pakistan
s.anwar@uettaxila.edu.pk

***Abstract*—We present a recommender system anticipated to be used by community of gamers. This system filters out or evaluate games through the opinions of other similar gamers using collaborative filtering technique and suggest those to the intended user. The system uses individual ratings given by the members of community, along-side rating of the games that a particular gamer likes, in order to predict and recommend new games to that gamer. The aim is to recommend games that match the user preferences and then user-based collaborative filtering is applied on the individual ratings of the games for a particular gamer to find similarity between those gamers. *Genre-based* filter is smeared on the shared rated games. The prediction algorithm is also tested on a larger standard dataset, Movie Lens, in order to verify the prediction accuracy. The results of our collaborative filtering approach is also compared using Mean Absolute Error. A working web based system is presented that found high user satisfaction in terms of usability and recommendation quality.**

Keywords*—*recommendation system, collaborative filtering, games genre

I. INTRODUCTION

Recommender systems are software that automatically predicts how much a user will like a particular item. Many top companies enhance customer satisfaction by investing in recommenders across the world as they bring real value to both, users and companies. Recommender systems are tools that interact with complex and large information spaces. They prioritize items likely to be of interest to the user and provide a personalized view of such spaces. The field, immersed in 1995, has grown hugely in the variety of problems addressed and techniques employed, as well as in its practical applications. Personalized recommendations are part of many e-commerce applications such as Netflix, Amazon and Pandora [1]. These recommendations based system have been missing in the video game industry [2].

Video and computer games have become more than a fad; it is a part of life. In the past, computer games have been considered as a distraction from more 'worthy' activities, such as homework or playing outside. Today, however, teachers and designers of learning resources and specially researchers are beginning to ask how this new medium might be powerful to support children's learning [3]. So, games are not only the source of entertainment but also help people learn and develop their mind to think better and also increases their problem solving ability. Video gameplay can improve certain aspects of cognitive ability [4]. Exergames: Digital games combining exercise with gameplay can provide social and academic benefits as well as improve youths' health status [5].

Recommendation systems are effective tools for personalization. We seek recommendations from friends and family since they know us well. Similarly, a recommendation system models this behavior by learning user preference to recommend things of interest [6]. Game recommender systems keeps gamers up-to-date with the latest games of their interest and help them to explore new games and excel in their passion. Collaborative filtering (CF) is one of the vigorous personalization technologies powering the adaptive web. [7].

CF try to predict the utility of items for an active user, based on the items rated by other users previously. More formally, the utility $u(c,s)$ of item s for user c is estimated based on the utilities $u(cj,s)$ assigned to item s by those users $cj \in C$ who are "similar" to user c. For example, in a game recommendation application, in order to recommend games to user c, the collaborative recommender system tries to find the "peers" of user c. Then, only the games that are most liked by the "peers" of user c would get recommended [8].

To make precise recommendations, the recommendation system must first learn the user's preferences from the ratings that the user makes. Also the new items that are added regularly to recommender system would not be recommend until the new item is rated by a substantial number of users. The proposed system aims to address these issues.

II. RELATED WORK

Text-reviews of the games were used for games recommendation to concerned users [9]. Pair of words were used containing adjectives and context words, phenomenon known as *information-theoretic co-clustering*, thus lessening the dimensionality of vectors. A precision of 0.84 – 0.86 was achieved using the co-clustering approach. Jose P. Zagal et. al. [10] emphasized on the usage of the term gameplay in user-submitted game reviews on various websites. They filtered the adjectives that altered gameplay using Natural Language Processing (NLP). After clustering the adjectives, they identified a common aesthetics of gameplay. Two things are depicted by this aesthetics, empirical support towards the significance of concepts necessary for understanding

978-1-4673-9073-6/17/$31.00 ©2017 IEEE

gameplay and indication of the variance of describing gameplay between players and designers/scholars.

Anthony Chow et.al. [11] focused on the recommendations of mobile games. They used personalized random-walk approach incorporated with both content-based and user-based approaches. This hybrid method was compared with customary collaborative filtering and content-based algorithms. G. Lee et.al.[12] introduced a sports commentary recommendation system (SCoReS) which can robotically suggest stories to viewers. They improved the enjoyment metrics by comparing their system with three other types of commentary hence added value to the watch-ability of sports.

A. ZIANI et.al. [13] used polarity detection and Arabic opinion analysis to propose recommender system for sports articles. Due to limited number of samples and large number of features, over-training occurred. Therefore, they merged the random sub space method and support vector machine classifier to elude over-fitting.

J. Herlockeret.al. [14] provided a set of recommendations to suggest design of neighborhood-based prediction systems using experimental study. To improve recommendation quality, collaborative tagging system is an interesting and beneficial technique to enhance the recommendation algorithms. Karen H. L et.al. [15] proposed a universal technique that allows tags to be fused to standard CF algorithms. BadrulSarwaret.al. [16] experimentally assessed a new algorithm for CF-based recommender systems showed that item-based techniques hold the promise of allowing CF-based algorithms to scale to large data sets and at the same time produce high-quality recommendations.

The proposed system is distinctive in such a way that not only a recommendation system is developed but also translated into an actual web-based system that maintains the accuracy and high predictive value. Game ratings gives us a better idea of the likeness or unlikeness of a game for a particular user, hence, these are used to find the predictions. This has overcome the drawback of unsupervised learning in which error-rate is much higher in recommending an item to specified user.

III. METHODOLOGY

The users of theproposed system were presented with two options i.e. general recommendations suggesting games according to interested genre and personalized recommendations that suggested games executing the whole recommendation algorithm.

The system first selects those games from the games data set that lie in active user's interested genre but still not rated by the user. It further filters this list of unrated games by finding similarity between unrated and highly rated games by the user using item-based collaboration and Pearson correlation. A subset of games that are most similar to the rated games from the list of unrated games will be passed to the recommender algorithm. It will employ user-based collaborative filtering and suggest games to the active user by finding a bunch of similar users and use their ratings to predict rating of a game for the active user. The data flow of the recommender system is shown in Figure 1.

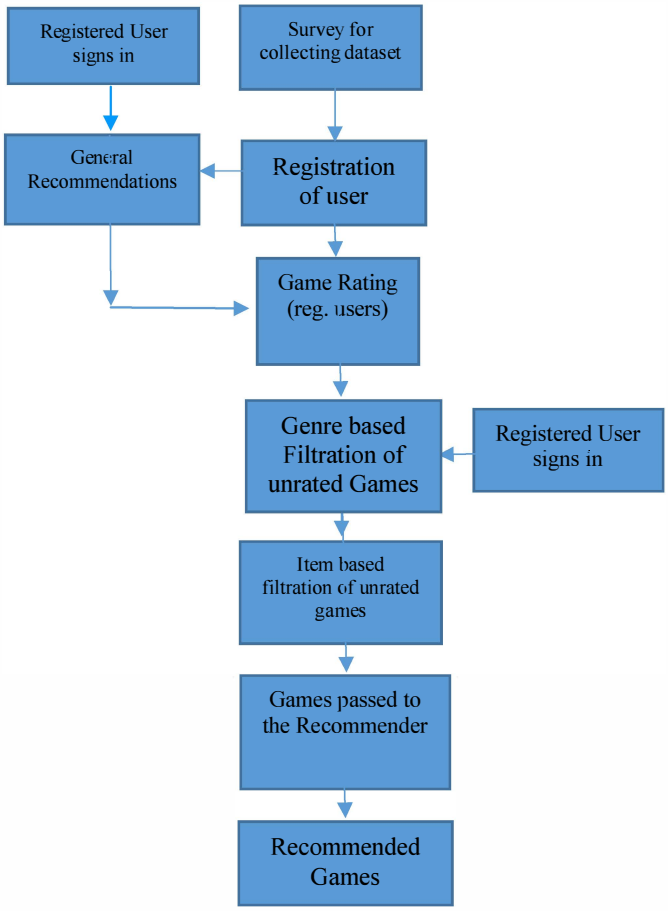

Figure 1. Flow diagram of the game recommender system

On the front end of the system, users register with an account on the website to get recommendations. During registration, the user is given two options,

1. General Recommendations
2. Personalized Recommendations (by Collaborative Filtering)

In case, 'personalized recommendations', is selected the user needs to rate at least three games from the provided genreof interest. This condition is enforced to avoid cold start scenario in collaborative filtering. In 'general recommendations', all the games that belong to user's interested genre and having accumulative rating above an average value are suggested to the user.

A. ITEM BASED FILTERING

Item based filtering is performed using *Pearson correlation* formula. Let us suppose that 'G' be the set of games, 'U' be the set of users who have rated the picked game.Let us say 'a' be the picked unrated game and 'b' represents rated game by active user, then the similarity between games 'a' and 'b'is calculated using,

$$Sim(G_a, G_b) = \frac{\sum(RG_a - AG_a)(RG_b - AG_b)}{\sqrt{\sum(RG_a - AG_a)^2 * \sum(RG_b - AG_b)^2}} \quad (1)$$

Where, RG_a and RG_b are the ratings of *Game* a and *Game* b respectively and $a, b \in G$, AG_a and AG_b be the average of *Game* a and *Game* b respectively and $a, b \in G$

After finding similarity of picked unrated game and all the rated games, selection of those rated games is made which are in the nearest neighbor (having maximum similarity with the unrated) of unrated picked game. Using these nearest neighbors, system predict rating of the picked unrated game for active user.

$$Prediction = \frac{\sum(Sim(G_a, G_b) * RG_b)}{\sum(|Sim|)} \quad (2)$$

Where, $\sum(|sim|)$ is the sum of mod of similarities between the picked unrated game and its nearest neighbors (rated games by active user) and RG_b be the rating of game 'b'by active user where 'b' $\in$ *group of rated games by active user*.

B. USER BASED FILTERING

The unrated games were sorted in descending order of prediction. The system then passes the list of unrated games having highest prediction to the User-based recommendation part of the algorithm.Similarity between the active user and other chosen gamers is then calculated using Pearson correlation formula.

Let us suppose, U be the set of chosen gamers/users, I be the set of games, 'i' be the gamer to whom system is going to recommend some game,

$$Sim(U_i, U_j) = \frac{\sum(RU_i - AU_i)(RU_j - AU_j)}{\sqrt{\sum(RU_i - AU_i)^2 * \sum(RU_j - AU_j)^2}} \quad (3)$$

Where, RU_i and RU_j are the ratings of gamer 'i' and gamer 'j' respectively, $where\ j \in U$, AU_i and AU_j be the average of gamer 'i' and gamer 'j' respectively and $where\ j \in U$. If the system predicts the rating of the chosen game greater than average 3.0 (since a scale was set for rating games from 1 – 5), then the game will be recommended to user/gamer, otherwise it will not be recommended. The next priority game from the list of unrated games will be picked and similar steps will be followed to either recommend or move on to the next game.

The system performance has been evaluated using these parameters [17],

True Positive Rate (TPR): $TPR = \frac{TP}{TP + FN} = \frac{TP}{P}$

False Positive Rate (FPR): $FPR = \frac{FP}{FP + TN} = \frac{FP}{N}$

Positive Predictive Value (PPV): $PPV = \frac{TP}{TP + FP} = \frac{TP}{P'}$

Accuracy: $ACC = \frac{TP + TN}{P + N}$

IV. RESUTLS AND DISCUSSION

A. DATA COLLECTION

Data was assembled and analyzed for 1200 ratings of 30 games in 5 different genres. The survey included 90 users, 55 of them were male and 35 were female. The mean ageof the participants was 22.3 years and reflect the age group that would demonstrate the highest interest in video games. The selected genres were sports, action, adventure, puzzle and first person shooting. The proposed algorithm for recommendations was also analyzed and validated on existing Movie Lens dataset [18]. This contains 1,000,209 ratings of approximately 3,900 movies rated by 6,040 Movie Lens users.

B. SYSTEM ANALYSIS

In order to assess the performance of proposed algorithm, a web-based system was developedthat was used to collect user data. The movie lens data was also used to validate the system performance. For the survey based data, the technique of cross-checking was used and a precision of 0.9 was achieved.

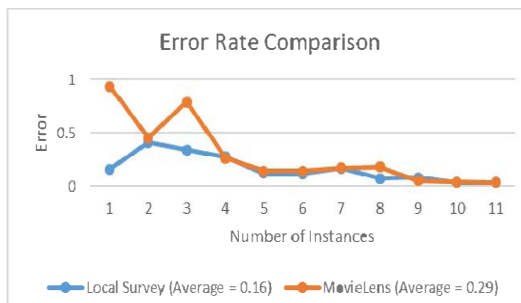

Figure 2. Error-rate comparison betweenlocal survey and MovieLens datasets

Moreover, along-with calculating the precision, error-rate was also measured by taking out the average difference of the actual rating given by gamer with the rating the proposed algorithm suggests. The system gave an average error-rate of 0.16. Movie Lens dataset was used since it is more vast and immense. The proposed algorithm achieved a precision of 0.83 on Movie Lens dataset.Error-rate was also calculated for Movie lens by taking out the average difference of the actual rating given by a Movie Lens user with the rating the proposed

algorithm suggests. An average error-rate of 0.29 was achieved.

Figure 2 shows the error-rate comparison of the algorithm for the two datasets. It is evident from the graph that the performance of the system improves with increasing number of tested instances. Table 1 shows the confusion matrix for the survey-based dataset. A total of 25 instances were checked against 30 games and 90 users. Out of these 25 occurrences, 21 were predicted correctly according to the actual value. The overall system performance parameters are as follows,

True Positive Rate (TPR) = **0.95**
False Positive Rate (FPR) = **0.3**
Positive Predictive Value (PPV) = **0.95**
Accuracy (ACC) = **0.92**

Table 1. Confusion matrix for the game rating dataset

	True Matches	**True Non-Matches**	
Predicted Matches	TP = 21	FP = 1	P` = 22
Predicted Non-Matches	FN = 1	TN = 2	N` = 3
			Total = 25

Table 2. Confusion matrix for movie lens dataset

	True Matches	**True Non-Matches**	
Predicted Matches	TP = 88	FP = 5	P` = 93
Predicted Non-Matches	FN = 4	TN = 3	N` = 7
			Total=100

Table 2 represents the confusion matrix for 100 random instances from 943 users on 1682 movies. Table 3presents a comparison of MAE (Mean Absolute Error). The proposed approach performed better than all other approaches. These results have been presented for the Movie Lens dataset which is quite huge as compared to our own survey dataset and hence verifies the efficacy of the proposed algorithm when scaled to larger data.

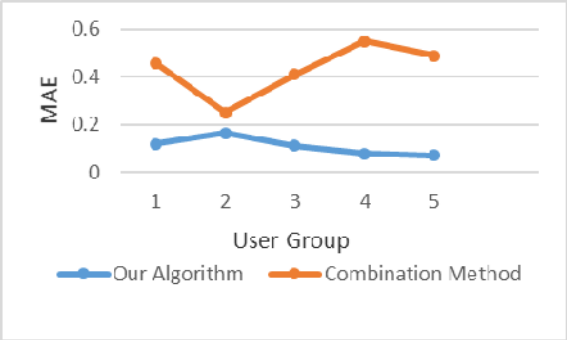

Figure 3: Comparison of the proposed method with CM method used for Movie Lens

A comparison of the MAE achieved using the proposed method for the movie lens data with [19] is shown in Fig. 3, which shows the improved performance.

Table 3: Comparison of MAE

Algorithm	**MAE**
Pure content-based (CB) predictor [12]	1.059
Naive Hybrid [11]	1.011
Content-boosted CF [17]	**0.962**
Proposed CF Algorithm	**0.29**

C. FRONT END DESIGN

A frontend website was developed using .NET framework to create a dynamic 3-in-1 platform. A single screen is shown in Fig. 4. To store huge amount of data, MS SQL Server 2012 was utilized. Proper web-semantics were used to ensure the organization of the web-interface. Moreover, bootstrap also came into play to make the website responsive and mobile-ready. This 3-in-1 platform facilitates its users to get to know about their favorite new and upcoming games on the homepage of the website. The users get download links and watch trailer of upcoming game. Secondly, a community like blog was also for gamers to discuss various games related issues like installation methods and other documentation. Alongside these added functionalities the core purpose of the system is to recommend games.

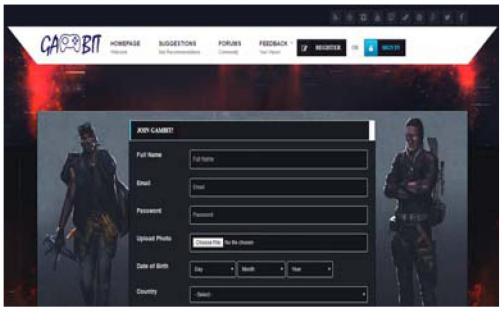

Figure 4: Frontend design of the web based system for game recommendation

V. CONCLUSION

A game recommender system is proposed that generates high quality recommendations to the users. The recommendation systems used by various websites have different efficiency levels depending on the context of the solution. The proposed recommendation system works efficiently according to games' data. The efficacy of the recommendation system has also been validated using the Movie lens data set. The growing video game industry can certainly benefit from this recommender system that would help in introducing the right games to the right audience and would be economically beneficial to the industry. # In future work, the approach can be extended in terms of game features used in order to find the similarity between users more efficiently.

ACKNOWLEDGMENT

We are highly grateful to the participants who rated games using our online system and also provided feedback for the system.

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